Custody Horizons for Ground-Based Angles-Only Tracking of Cislunar Periodic Orbits

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Abstract

Ground-based telescopes lose cislunar objects for days at a time. This paper quantifies the custody horizon: how long an object on a cislunar periodic orbit can go unobserved before it leaves a telescope search pattern centred on the operator's own unscented-transform prediction. Four Earth–Moon orbits (the 9:2 near-rectilinear halo orbit, an $\boldsymbol{L_1}$ halo, an $\boldsymbol{L_2}$ Lyapunov orbit and a distant retrograde orbit) are observed for one year by a simulated four-site network of 1 m telescopes. The targets stay within 14° of the Moon, so the Moon-exclusion angle matters more than aperture, and the longest lunar-phase blackout lasts 8.5–15.7 days. After a 3-day gap on the $\boldsymbol{L_1}$ halo, the extended Kalman filter reacquires consistently in 37 % of 100 runs, against 100 % for a Gaussian-mixture unscented filter. The operator horizon equals that of an idealised search centred on the true mean (median ratio 1.00). The near-rectilinear and distant retrograde orbits stay in custody beyond 30 days after three or more days of tracking, whereas the unstable $\boldsymbol{L_1}$ and $\boldsymbol{L_2}$ orbits are lost after 11–17 days (median, ten fields), depending mainly on the tracking-arc length. Across 30 members of five families, the horizon falls with the stability index (Spearman $\boldsymbol{\rho = -0.92}$ above index 10). Individual horizons are predicted by the unscented-transform covariance, not by the finite-time Lyapunov exponent. Site biases and solar-pressure errors shorten the horizons by about 1 % at nominal levels but make covariance-based custody claims overconfident. The main conclusions are reproduced for 60 cases in a DE440 ephemeris model with solar radiation pressure.

Keywords: Cislunar space domain awareness, Angles-only orbit determination, Near-rectilinear halo orbit, Uncertainty propagation, Gaussian mixture filter, Custody

1 Introduction

Traffic beyond geosynchronous orbit is growing: lunar orbiters, the Gateway's 9:2 near-rectilinear halo orbit (NRHO), and spacecraft in libration-point and distant retrograde orbits (DROs). Keeping track of these objects, that is, maintaining *custody*, is harder than in near-Earth space for three reasons. The targets are faint and about 380 000 km away. Seen from Earth, they sit close to the brightest object in the night sky. And their motion is governed by the Earth–Moon three-body problem, which stretches uncertainty rapidly and makes it non-Gaussian [1–3].

Ground-based optical telescopes are the most accessible sensors for this task. Sechrest et al. [4] detected CAPSTONE at $g \approx 19.6$ and Orion at $g \approx 15.5$ with 36 cm telescopes. But ground sensors see cislunar objects only intermittently: daylight, the Moon's glare and the lunar phase open and close observation windows. Earlier work has shown that filters survive or fail across particular gaps on particular trajectories [5–7]. What is missing is a systematic answer, at least for representative orbits, to the operational question: *how long can a cislunar object go unobserved before it cannot be found again, and what determines that time?*

This paper answers that question for ground-based, angles-only tracking of periodic orbits in a simulation study. Its contributions are:

1. A physically generated observation model (sites, twilight, Moon exclusion and occultation, shadows, photometry with scattered moonlight). It shows that the targets studied stay within 14° of the Moon and that monthly lunar-phase blackouts of up to two weeks cannot be removed by adding ground sites (Section 5).
2. A definition of the *custody horizon* T_c as the longest gap after which 99 % of the true probability mass still lies inside a telescope search pattern, and maps of T_c across orbits, tracking-arc lengths and search sizes for a search centred on the operator's own unscented prediction, which keeps custody as long as an idealised search centred on the truth (Section 7).
3. Evidence that T_c can be predicted without Monte Carlo from the propagated post-fit covariance (unscented transform), but not from the orbit's finite-time Lyapunov exponent alone (Section 8).
4. A quantification of when the filter choice matters: linear (extended Kalman filter) prediction gives false custody on unstable orbits after long gaps, and a Gaussian-mixture filter reacquires far more reliably than the extended and unscented Kalman filters after a gap (Sections 6 and 8).
5. A cross-check in a DE440 ephemeris model with solar radiation pressure, using multiple-shooting ephemeris counterparts of the orbits (Section 9).
6. A sweep over 30 members of five orbit families, showing that nearly stable members keep custody beyond 30 days and that, above a stability index of 10, the horizon falls with the index (Section 10).
7. A consider-covariance analysis with site biases, clock offsets and solar-pressure errors, showing that the horizons are robust to these errors but that custody claims based on the Cramér–Rao covariance are not (Section 11).

2 Related Work

Surveys of the cislunar problem [2, 3] describe the growing traffic, the dynamical environment and the sensing challenges, among them faint targets close to the Moon and long observation gaps. Frueh et al. [1] showed that even a global ground network leaves periods with no visibility because of lunar brightness, and that cislunar uncertainty develops strongly curved, non-Gaussian structure.

Much of the work on observation geometry concerns space-based observers. Vendl and Holzinger [8] identified cislunar periodic orbits that give persistent detection capability, Fowler and Paley [9] derived observability metrics for space-based cislunar observers, Wilmer and Bettinger [10] studied NRHO surveillance from cislunar periodic orbits, and Bhadauria et al. [11] optimised the surveillance of key cislunar regions. From the ground, Sechrest et al. [4] demonstrated detections of CAPSTONE and Orion, and Evans et al. [12] combined lunar-phase and Earth-rotation visibility constraints with an information-matrix covariance analysis of angles-only tracking of six periodic-orbit families from a single ground station. That analysis, like Section 6 here, quantifies the uncertainty at the end of tracking; the present work continues from that uncertainty through the observation gap to the search that must find the object again.

Custody across gaps has so far been studied with particular filters on particular trajectories. Chow et al. [5, 6] processed ground-based angles-only and photometric observations from three sites for 15 periodic orbits. They found that Gaussian-mixture filters retain custody across data gaps where an unscented Kalman filter fails, and that two-body models lose the object after the first gap. Their gaps, however, were produced by decimating the data rather than by physical visibility. For space-based observers, Iannamorelli and LeGrand [7, 13] kept custody of a manoeuvring Artemis-like target across gaps of 1.7 and 2.6 days with an adaptive Gaussian-mixture interacting-multiple-model filter, and Greaves and Scheeres [14] estimated and detected manoeuvres of cislunar vehicles from optical measurements with an optimal-control-based estimator. For initial orbit determination, Jo et al. [15] developed a three-body differential-correction method for angles-only data and reviewed the alternatives.

On the estimation side, uncertainty propagation in orbital mechanics is reviewed by Luo and Yang [16]. For strongly nonlinear dynamics, state transition tensors [17], the unscented transform [18, 19], Gaussian mixtures with entropy-triggered splitting [20] and regularised particle filters with progressive correction [21, 22] trade cost against fidelity; the last three are used here. Consistency is assessed with the normalised estimation error squared (NEES) [23]. Whether a predicted covariance is realistic, and not only consistent with the filter's own model, is the subject of covariance-realism work in space surveillance [24]; the classical tool for errors that the estimator does not model is the consider covariance [25], used in Section 11. The dynamical tools are standard methods for the circular restricted three-body problem [26, 27]: Richardson's third-order halo approximation [28], differential correction and pseudo-arclength continuation. The finite-time Lyapunov exponent (FTLE) [29] is the natural candidate for predicting how fast knowledge decays, and it is tested directly here.

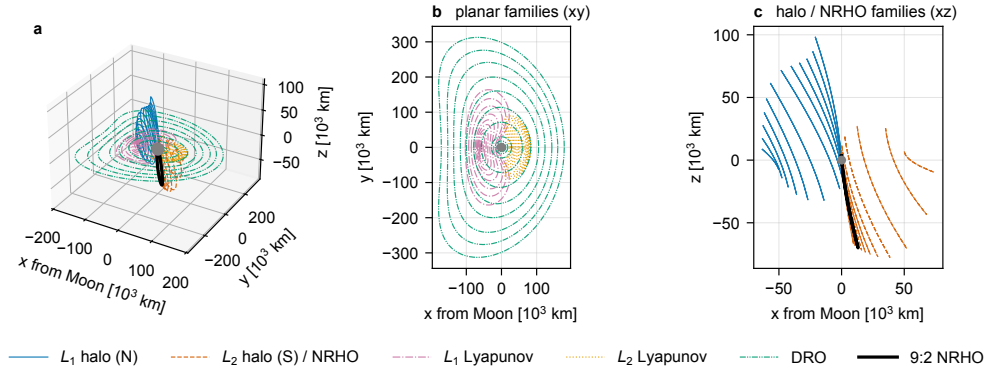


Fig. 1 Earth–Moon CR3BP families used in this study (12 members of each family shown), in Moon-centred coordinates: **a** three-dimensional view, **b** planar families in the xy plane, **c** halo and NRHO families in the xz plane. The thick black curve is the 9:2 southern NRHO

3 Models

3.1 Dynamics and orbit catalogue

Motion is modelled in the Earth–Moon circular restricted three-body problem (CR3BP), in the barycentric rotating frame with the Earth at $(-\mu, 0, 0)$ and the Moon at $(1-\mu, 0, 0)$:

$$\ddot{x} - 2\dot{y} = \frac{\partial \Omega}{\partial x}, \quad \ddot{y} + 2\dot{x} = \frac{\partial \Omega}{\partial y}, \quad \ddot{z} = \frac{\partial \Omega}{\partial z}, \qquad \Omega = \tfrac{1}{2}(x^2 + y^2) + \frac{1-\mu}{r_1} + \frac{\mu}{r_2}, \tag{1}$$

with $\mu = 0.0121505842$ (from DE440 gravitational parameters), a length unit of 384 400 km and a time unit of 4.3425 days. The state transition matrix (STM) $\Phi(t, t_0)$ is integrated with the variational equations using an eighth-order Runge–Kutta method [30, 31]; it agrees with Richardson-extrapolated finite differences to 3×10^{-10}.

Symmetric periodic orbits are computed by single-shooting differential correction and continued by pseudo-arclength continuation, with halo families seeded by Richardson's third-order solution. Five families are built: the L_1 northern halo (88), the L_2 southern halo down to the NRHOs (140), the L_1 and L_2 Lyapunov families (60 each) and the DRO family (94); the numbers are family members (Fig. 1). The 9:2 synodic-resonant NRHO is found by period targeting. It has a period of 6.5624 d, perilune and apolune radii of 3249 and 71 222 km, and a stability index of 1.32, consistent with Zimovan-Spreen et al. [32]. Four representative orbits are used throughout: the 9:2 NRHO, an L_1 halo with out-of-plane amplitude $A_z \approx 30\,000$ km, a mid-family L_2 Lyapunov orbit, and a DRO whose closest approach to the Moon is about 70 000 km.

3.2 Sky geometry

To give the sky geometry real dates, the CR3BP trajectory is placed in the instantaneous Earth–Moon frame of the actual Moon. The frame's $\hat{x}$ axis points from Earth to Moon, its $\hat{z}$ axis lies along the Moon's orbital angular momentum, and distances scale

with the constant length unit. The Sun and Moon come from low-precision analytic series [33], and Earth rotation from Greenwich mean sidereal time. The Moon used in every geometric test is the mapped CR3BP Moon, so the target–Moon geometry is exactly the CR3BP one. Truth, filters and predictors share this mapping, so the setup is self-consistent; Section 9 replaces it with DE440 [34]. The reference epoch is 2027-01-01 00:00 UTC and the time grid spacing is 10 min.

3.3 Sensors and visibility

The network consists of Hanle (India), La Palma (Spain), Haleakala (USA) and Siding Spring (Australia), in subsets of one site (India-1), three sites (Tri-3) and four sites (Tri-3+S). An observation is possible only if all of the following hold: the Sun is below $-18°$; the target is above $20°$ elevation; the line of sight misses the Moon; the target is outside Earth's and the Moon's (cylindrical) shadows; the target is more than θ_{m} from the Moon; and the target is brighter than the effective limiting magnitude. The target is a Lambertian sphere of radius 1 m (2 m diameter) and albedo 0.2, of magnitude 18.4 at zero phase angle at the Moon's distance. The limiting magnitude is degraded by scattered moonlight following Krisciunas and Schaefer [35], and atmospheric extinction is applied. Dark-sky limiting magnitudes of 19.5, 21.5 and 23.0 are assumed for 0.36, 1 and 2 m apertures; the 0.36 m value is anchored to the CAPSTONE detection of Sechrest et al. [4]. Measurements are topocentric right ascension and declination with $1''$ noise in each axis on the sky (isotropic, so the right-ascension noise is $1''/\cos\delta$), every 10 min per site when the target is visible.

3.4 Estimators

Four filters share one measurement model:

- an extended Kalman filter (EKF), with STM covariance propagation and Joseph-form updates;
- an unscented Kalman filter (UKF) with 13 scaled sigma points ($\alpha = 1$, $\beta = 2$, $\kappa = 0$);
- a Gaussian-mixture UKF (GM-UKF). A component is split when its unscented prediction raises the differential entropy of its Gaussian fit by more than 0.02 nats [20]. Because the flow preserves volume, any entropy gain comes from nonlinear bending. The split is a moment-preserving three-way split along the direction of maximum stretching (the top right-singular vector of the statistically linearised flow multiplied by the Cholesky factor of the covariance). Pruning and Mahalanobis merging keep the mixture at no more than 40 components;
- a regularised bootstrap particle filter (PF) with progressive correction [21, 22] and 3000 particles. It hands over to a UKF once its position uncertainty falls below 50 km. Without the handover, the finite particle set impoverishes on long, dense arcs and its covariance drops below the Cramér–Rao bound.

The truth has no process noise. The filters carry a small white acceleration noise with power spectral density $10^{-18}\,\mathrm{km^2\,s^{-3}}$.

3.5 Declaration of AI-assisted tools

The author used Claude (Anthropic), a large language model, as an assistant for writing and editing code and manuscript text, for checking the integrity of the code and fixing bugs, and for checking references. The author conceived and directed the study, chose the models, parameters and error budgets, ran all simulations and interpreted the results, and reviewed and verified all AI-assisted code and text, including through the unit tests and an automated check of every number in the paper against the generated data. The author takes full responsibility for the content of this manuscript.

4 Custody Horizon: Definition and Computation

Let $(\hat{\boldsymbol{x}}_0, P_0)$ be the post-fit estimate at the last measurement before a gap. For gap lengths $\Delta t \in [0.1, 30]$ d (120 logarithmically spaced values), the uncertainty is propagated three ways:

- *linear*: $P(\Delta t) = \Phi P_0 \Phi^\top$, which is what an EKF predicts;
- *unscented* (UT): 13 sigma points, which is what a UKF predicts;
- *Monte Carlo* (MC): 300 samples of $\mathcal{N}(\hat{\boldsymbol{x}}_0, P_0)$ through the full dynamics, used as the reference.

The samples are projected onto the geocentric sky. For a search pattern of N square fields of view of side ϕ, whose equal-area radius is $r_N = \phi\sqrt{N/\pi}$, the custody horizon is the first gap at which less than 99 % of the probability mass lies inside the search circle,

$$T_\mathrm{c}(N) = \min\left\{\Delta t : \Pr\left[\angle(\hat{\boldsymbol{u}}_\mathrm{target}, \hat{\boldsymbol{c}}) \le r_N\right] < 0.99\right\}, \tag{2}$$

where $\hat{\boldsymbol{c}}$ is the centre of the search. Three variants use different centres or radii:

- *ideal*: the circle is centred on the true (MC) mean direction;
- *actual*: the circle is centred on a method's own predicted direction;
- *claimed*: the time at which a method's *own* 99 % region outgrows r_N.

The *operator horizon* is the actual horizon of the UT prediction: the circle is centred where a UKF would point the telescopes, so it uses no knowledge of the truth. It is the horizon reported in Tables 3 and 4; the ideal horizon is the reference for the predictors of Section 8. A claimed horizon longer than the actual one is *false custody*: the predictor believes the target is still inside the pattern after it has left. Here $\phi = 1°$ and $N \in \{1, 10, 100\}$, giving search radii of 0.56°, 1.78° and 5.64°. Gaussian *containment* is the share of MC samples inside a method's 99 % sky ellipse (ideal value 99 %).

Because T_c is the *first* gap at which the 99 % radius exceeds r_N, the gap grid must resolve short excursions of the sky radius. On the NRHO the radius spikes at every perilune passage (every 6.56 d), when the along-track uncertainty is stretched as the object sweeps past the Moon, and then shrinks again. The horizon is interpolated between grid points in log–log coordinates. For 17 representative arcs, refining the grid to a uniform 0.02-d step changes T_c by a median of at most 0.03 d, by at most 0.15 d for $N = 1$ and 10, and by at most 0.17 d for $N = 100$.

Table 1 One year of visibility (1 m telescopes, Tri-3+S network): share of time observable and longest observation gap, by Moon-exclusion angle θ_{m}

	$\theta_{\mathrm{m}} = 2°$		$\theta_{\mathrm{m}} = 5°$		$\theta_{\mathrm{m}} = 10°$	
Orbit	Visible	Max gap	Visible	Max gap	Visible	Max gap
9:2 NRHO	37.5 %	12.8 d	31.4 %	13.2 d	4.5 %	19.5 d
L_1 halo	42.7 %	11.2 d	32.6 %	11.8 d	0 %	—
L_2 Lyapunov	32.7 %	13.5 d	20.6 %	14.5 d	0 %	—
DRO	38.0 %	11.8 d	32.5 %	11.9 d	21.2 %	16.5 d

The post-fit covariance at the gap start is the Cramér–Rao lower bound (CRLB) of the preceding tracking arc of 1, 3 or 7 days:

$$\mathcal{I}(t_0) = \Lambda_0 + \sum_k \Phi(t_k, t_0)^\top H_k^\top R_k^{-1} H_k \Phi(t_k, t_0), \qquad P_0 = \Phi(t_e, t_0)\, \mathcal{I}^{-1}\, \Phi(t_e, t_0)^\top, \quad (3)$$

built from the actual visible measurements, with the gap starting at the arc's last measurement t_e. Here H_k is the Jacobian of right ascension and declination, R_k the measurement noise covariance ($\mathrm{diag}(\sigma^2/\cos^2\delta_k, \sigma^2)$ with $\sigma = 1''$, the same model as in the simulated measurements and the filters), and Λ_0 a weak regularising prior (10^5 km, $1\,\mathrm{km\,s^{-1}}$). An arc is counted as observable when the resulting position bound is finite and below this prior. Section 6 shows that the filters reach this bound in tracking. Two scenarios are studied: arcs ending every 15 days through the year (*phase*), and arcs ending exactly when a blackout of 2 days or more begins (*blackout*). Together they give 523 cases, of which 393 are distinct arcs: a phase arc often ends exactly where a blackout begins, and then appears in both scenarios. Statistics pooled over both scenarios use each distinct arc once.

A purely dynamical predictor is also tested: $\hat{T}_{\mathrm{FTLE}}$ is the first Δt at which the largest singular value of Φ, $s_{\max}(\Phi(\Delta t)) = e^{\sigma_{\Delta t}\Delta t}$, reaches r_N/θ_0, where θ_0 is the initial 99 % sky radius and $\sigma_{\Delta t}$ is the FTLE [29].

5 Visibility: the Moon and the Lunar Phase Dominate

Seen from Earth, the four targets stay close to the Moon (Fig. 2). The L_1 halo stays within 3.9–6.1° and the L_2 Lyapunov orbit within 0–7.0°. The NRHO ranges over 0.5–10.0° (median 8.3°), because it spends most of its time near apolune, and the DRO over 0–13.9°. All targets therefore stay within 14° of the Moon, and the Moon-exclusion angle is the dominant sensor parameter (Table 1). With a 10° exclusion, the L_1 and L_2 targets are never observable, whereas at 5° they are observable 21–33 % of the year. At 5°, doubling the aperture from 1 to 2 m changes the network's visible share by only 0.5–1.7 percentage points.

Visibility arrives in windows that run from roughly first quarter to last quarter (Fig. 3): near new Moon the target lies close to the Sun on the sky and is up only in daylight, and near full Moon the exclusion angle and scattered moonlight bite.

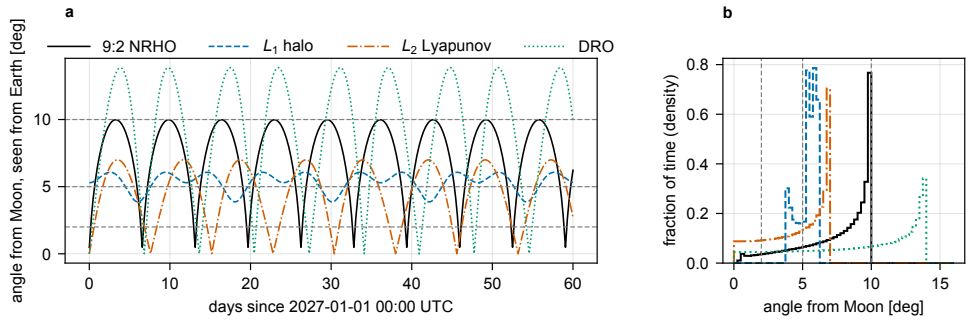


Fig. 2 Geocentric angular separation between each target and the Moon: **a** 60-day time series, **b** distribution over one year. Dashed lines mark the exclusion angles studied

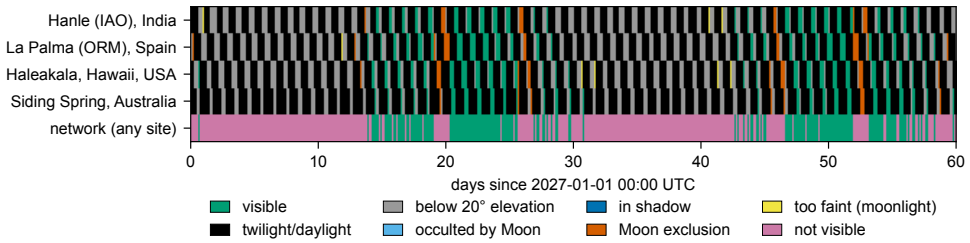


Fig. 3 First blocking constraint per site for the 9:2 NRHO over 60 days (1 m telescopes, 5° exclusion). The bottom row shows network availability

Gap lengths fall into three regimes (Fig. 4): handovers between sites (a few hours), diurnal gaps of about 1 d, which dominate for a single site, and multi-day lunar-phase blackouts. Adding sites or aperture removes short gaps but not the lunar-phase blackout: at 5° exclusion, the longest gap of the year is 8.5–15.7 d for every network, 1–2 m aperture and orbit.

6 Observability and Filter Performance

6.1 Cramér–Rao bounds

Figure 5 shows the bound, computed from the actual measurement schedule, for arcs starting every 2 days. At the end of 7-day arcs the median worst-direction position uncertainty is 1.14 km (NRHO), 1.83 km (L_1 halo), 7.03 km (DRO) and 17.8 km (L_2 Lyapunov), and the median velocity uncertainty is 6–108 mm s^{-1}. Between 75 % and 84 % of 7-day arcs are observable. For the DRO and the Lyapunov orbit, the weakest direction lies within 6.4–7.8° (median) of the line of sight, that is, it is essentially range. For the NRHO and the halo it lies further from the line of sight (median 12.8–17.9°), because the strongly curved dynamics over a week also constrain range. Arcs that end inside a blackout show the uncertainty growth that motivates the rest of the paper.

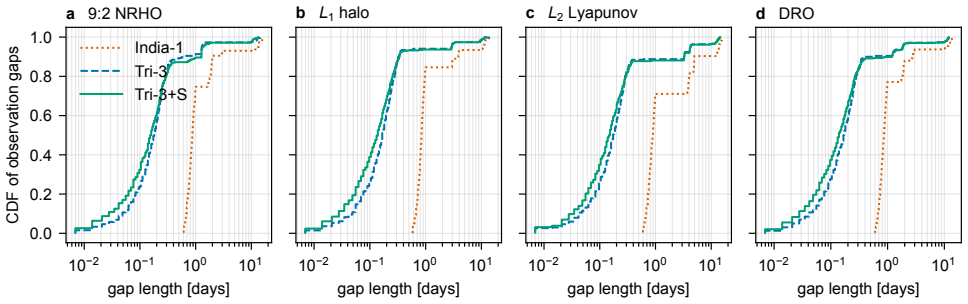


Fig. 4 Cumulative distribution of observation gaps over one year (1 m telescopes, 5° exclusion) for the three networks: **a** 9:2 NRHO, **b** L_1 halo, **c** L_2 Lyapunov, **d** DRO

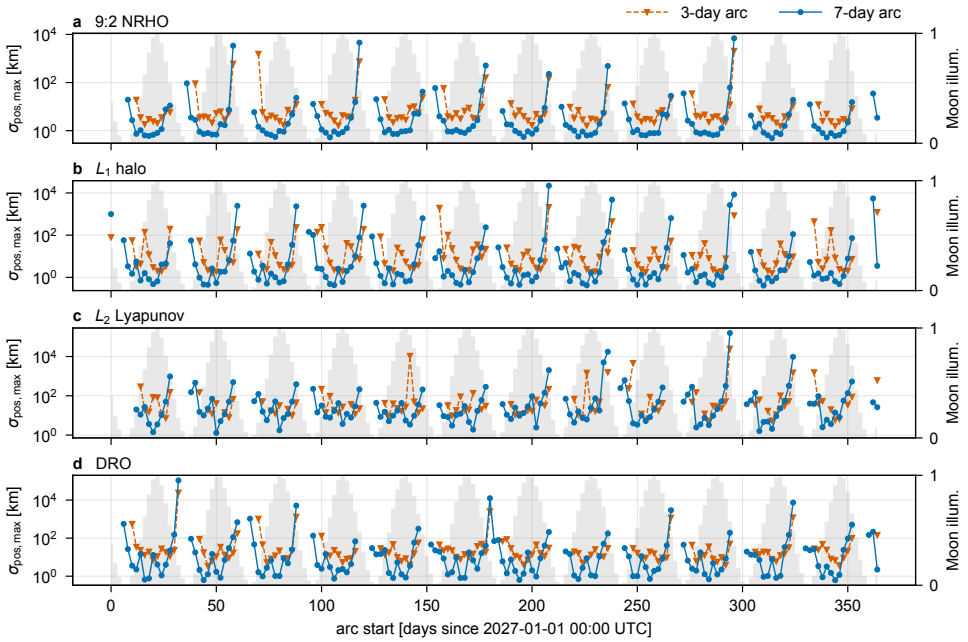


Fig. 5 Cramér–Rao bound on the worst-direction position uncertainty at the end of 3-day and 7-day arcs, against the arc start date: **a** 9:2 NRHO, **b** L_1 halo, **c** L_2 Lyapunov, **d** DRO. Grey shading shows lunar illumination; gaps are arcs that are not observable

6.2 Tracking

When the filters are started at the first measurement of the arc from a 1000 km / 10 m s^{-1} (1σ) prior, both the EKF and the UKF converge to the bound (Table 2, 20 Monte Carlo runs per orbit). The averaged NEES (ANEES, ideal value 6) lies inside the 95 % band for 20 runs (4.58–7.61) for every orbit and both filters except the EKF

Table 2 Filter performance on 7-day arcs. Tracking: 20 runs, prior at the first measurement. Reacquisition (L_1 halo, arc starting at a 3.0 d observation gap): 100 runs, prior at the arc start; brackets give 95 % Wilson intervals; run times are wall-clock values on a laptop and machine dependent; the PF values vary slightly between platforms because resampling amplifies rounding differences. CRLB and errors are position values in km (errors: RMS for tracking, median and maximum for reacquisition). "Consistent" means the final NEES is below $\chi^2_{0.99}(6)$

Tracking					
Orbit	CRLB	EKF error	EKF ANEES	UKF error	UKF ANEES
9:2 NRHO	1.16	1.58	5.13	1.58	5.14
L_1 halo	1.83	1.94	5.71	1.97	5.36
L_2 Lyapunov	17.88	9.60	9.25	9.50	6.05
DRO	7.16	6.05	5.13	6.15	5.15
Reacquisition, L_1 halo					
Filter	Consistent	Median error	Max error	ANEES	Time [s]
EKF	37 % (28–47)	68.0	1792.7	3.4×10^3	0.04
UKF	77 % (68–84)	31.0	251.5	64.0	0.06
GM-UKF	100 % (96–100)	24.3	95.4	5.9	0.11
PF→UKF	88 % (80–93)	27.6	1546.9	1.3×10^3	0.19

on the L_2 Lyapunov orbit, which is overconfident (9.25); the UKF on the same arc is consistent (6.05).

6.3 Reacquisition after a gap

Reacquisition is tested on a separate 7-day arc per orbit that starts at the onset of the observation gap closest to 3 days, with the prior placed at the arc start. In the L_1 halo arc the first measurement arrives 3.0 d after the prior epoch. By then the 1000 km prior has stretched into a cloud whose 1–99 % extent is about 42 000 × 7600 km (Fig. 6). Over 100 runs, only 37 % of EKF runs end consistent (95 % Wilson interval 28–47 %), with errors of up to 1793 km while the filter claims about 36 km. The UKF reaches 77 % (68–84 %), the PF with UKF handover 88 % (80–93 %), and the GM-UKF 100 % (96–100 %) (Table 2, Fig. 7). The GM-UKF splits into up to 27 components along the stretching direction, averaging 3.0 components over the arc. All differences between the GM-UKF and the other three filters are significant (Fisher exact test, $p \leq 0.02$). On the other three orbits, with gaps of 1.9–3.2 d, the UKF and the GM-UKF end consistent in 94–99 % of runs, the PF→UKF in 91–95 % and the EKF in only 60–85 %. At the first measurement on the L_1 halo, the EKF's 99 % ellipsoid contains only 73 % of the particles, against 94 % for the UKF and 96 % for the GM-UKF. After a 3-day gap the linear *prediction* therefore already misses part of the probability mass, and the measurement update, which linearises the angles across a 10^4 km cloud, adds to the error.

7 Custody Horizons

7.1 Horizon maps

Figure 8 shows how the true 99 % sky radius grows after 7-day arcs. The NRHO and DRO stay below 0.01° for 30 days (median), reflecting their near-stability. The

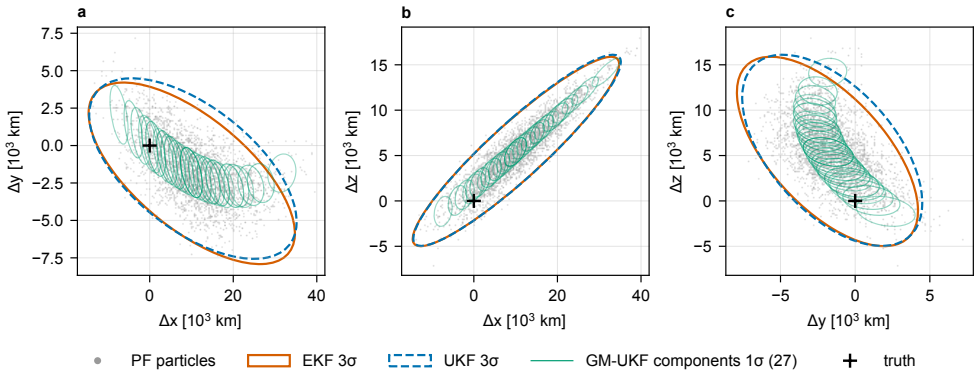


Fig. 6 L_1 halo: predicted uncertainty at the first measurement after the 3.0-day initial gap (run 0), relative to the truth, in the **a** xy, **b** xz and **c** yz synodic planes. Grey dots are PF particles; EKF and UKF 3σ ellipses and GM-UKF components (1σ) are shown

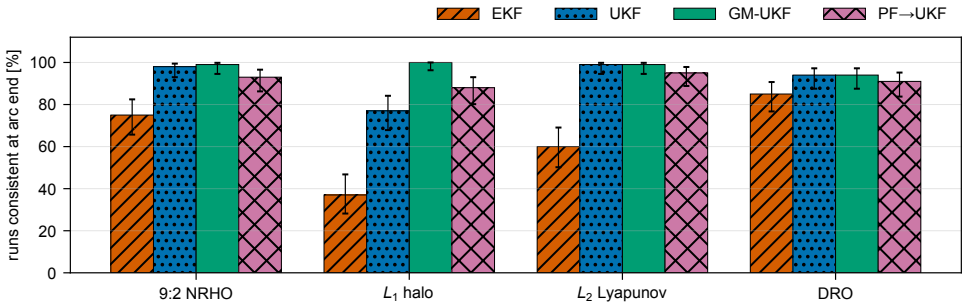


Fig. 7 Share of reacquisition runs ending consistent, by filter and orbit (100 runs each; initial gaps of 1.9–3.2 d); error bars are 95 % Wilson intervals

L_1 halo and L_2 Lyapunov orbits grow roughly exponentially after about 5 days, and their median operator horizons after 7-day arcs lie at 13.7–19.1 d for one to 100 fields. Because the growth is so steep, enlarging the search from 10 to 100 fields gains only 1.7–3.9 days (Table 3). The length of the tracking arc matters more: the L_1 halo's median horizon rises from 10.9 to 16.4 d (10 fields) as the arc grows from 1 to 7 days. Over all arcs, the unstable orbits' median horizons for a ten-field search lie at 10.9–17.3 d.

The horizons in Tables 3 and 4 are operator horizons: the search is centred on the UT-predicted direction. Over all 393 distinct arcs they agree with the ideal horizons (search centred on the true mean) to within 0.8 d for $N \leq 10$ and 1.5 d for $N = 100$, with a median ratio 1.00 for all three search sizes. When custody is lost, the UT-predicted centre is therefore still much closer to the true mean than the search radius; custody ends because the distribution outgrows the search, not because the search is centred in the wrong place.

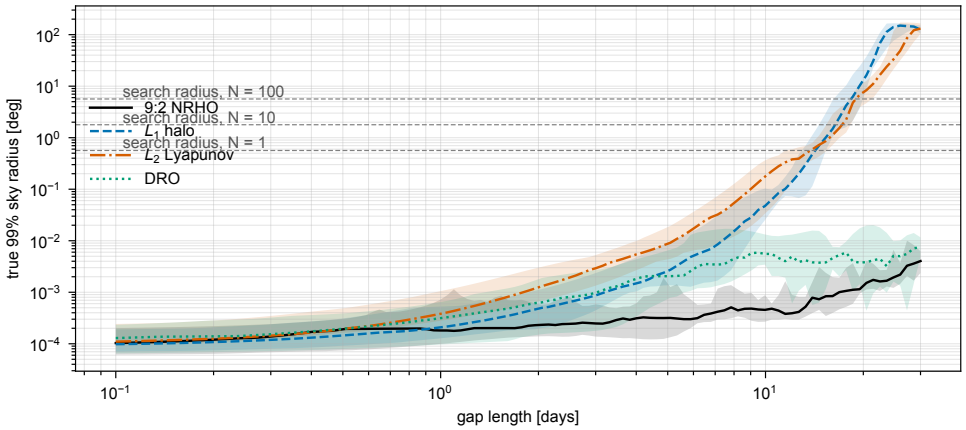


Fig. 8 Growth of the true (Monte Carlo) 99 % sky radius after 7-day arcs: median and 10–90 % band over the year. Dashed grey lines are the search radii for 1, 10 and 100 one-degree fields

Table 3 Custody horizon T_c [days], median (10th–90th percentile) over arcs through the year (phase scenario, operator horizon: search centred on the UT prediction); σ_0 is the median worst-direction position uncertainty at the gap start. ">30" means beyond the 30-day grid

Orbit	Arc	σ_0 [km]	$N = 1$	$N = 10$	$N = 100$
9:2 NRHO	1 d	24.5	>30 (5.3–>30)	>30 (18.4–>30)	>30
	3, 7 d	4.1, 0.9	>30	>30	>30
L_1 halo	1 d	20.8	8.8 (5.5–9.9)	10.9 (7.3–12.2)	12.6 (9.5–13.7)
	3 d	5.1	11.6 (7.9–12.4)	13.2 (10.0–14.1)	14.9 (13.4–17.1)
	7 d	0.8	14.3 (13.0–15.3)	16.4 (14.6–18.3)	18.3 (16.3–19.9)
L_2 Lyapunov	1 d	33.5	9.4 (7.6–10.9)	12.3 (9.7–15.4)	16.2 (12.8–18.0)
	3 d	16.2	10.3 (9.0–13.4)	14.9 (11.2–16.4)	18.0 (15.3–18.8)
	7 d	7.4	13.7 (11.6–15.7)	17.3 (15.4–18.3)	19.1 (18.3–21.1)
DRO	1 d	31.9	>30 (7.8–>30)	>30	>30
	3, 7 d	11.7, 1.4	>30	>30	>30

7.2 Blackouts

The operational question is whether custody survives the blackouts that actually occur. Figure 9 and Table 4 compare T_c with every blackout of 2 days or more in the year. After 7-day arcs, every blackout is survived with a ten-field search. After 1-day arcs, 11–24 % of the blackouts are lost for the L_1 and L_2 targets, and 22 % for the NRHO, whose blackouts last about 10.5 d. The number of blackouts per orbit is small (9–34), so the individual shares are uncertain by 10–30 percentage points (95 % Wilson intervals); the ordering by arc length, however, holds for every orbit.

Table 4 Blackouts (≥2 d) survived, that is, operator $T_c \geq$ blackout length, by tracking-arc length; n is the number of blackouts with an observable preceding arc. The ideal horizon gives the same shares except in one cell (L_1 halo, 1-day arcs, $N = 1$: 63 %)

Orbit	Arc	n	Median length	$N = 1$	$N = 10$	$N = 100$
9:2 NRHO	1 d	9	10.5 d	67 %	78 %	78 %
	3 d	11	10.5 d	100 %	100 %	100 %
	7 d	11	10.5 d	100 %	100 %	100 %
L_1 halo	1 d	27	3.1 d	67 %	89 %	93 %
	3 d	27	3.1 d	93 %	96 %	96 %
	7 d	28	3.1 d	100 %	100 %	100 %
L_2 Lyapunov	1 d	33	4.1 d	67 %	76 %	94 %
	3 d	34	4.1 d	79 %	97 %	97 %
	7 d	34	4.1 d	88 %	100 %	100 %
DRO	1 d	13	11.3 d	46 %	85 %	85 %
	3 d	13	11.3 d	100 %	100 %	100 %
	7 d	13	11.3 d	100 %	100 %	100 %

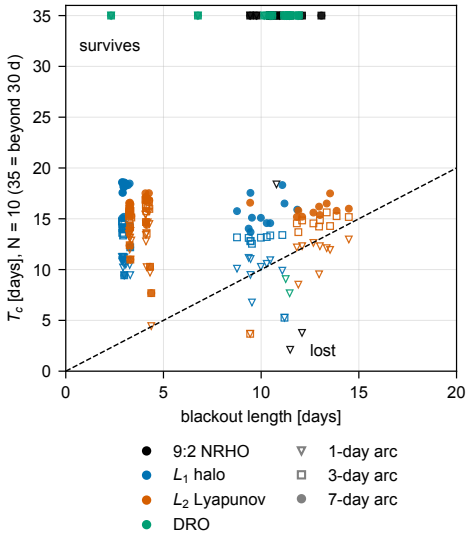


Fig. 9 Ideal custody horizon (ten fields; the operator horizon differs by at most 0.8 d) against blackout length. Points above the diagonal survive the blackout; marker shape shows the tracking-arc length (open: 1 and 3 days, filled: 7 days)

8 Predicting the Custody Horizon, and False Custody

Can T_c be predicted without Monte Carlo? Figure 10 compares the Monte Carlo horizon with two cheap predictors, using the 250–270 distinct arcs (depending on the predictor and N) whose horizons are finite and below 30 d; R^2 is computed on the logarithm of the horizon. The *orbit-only* FTLE predictor fails, with log-R^2 between

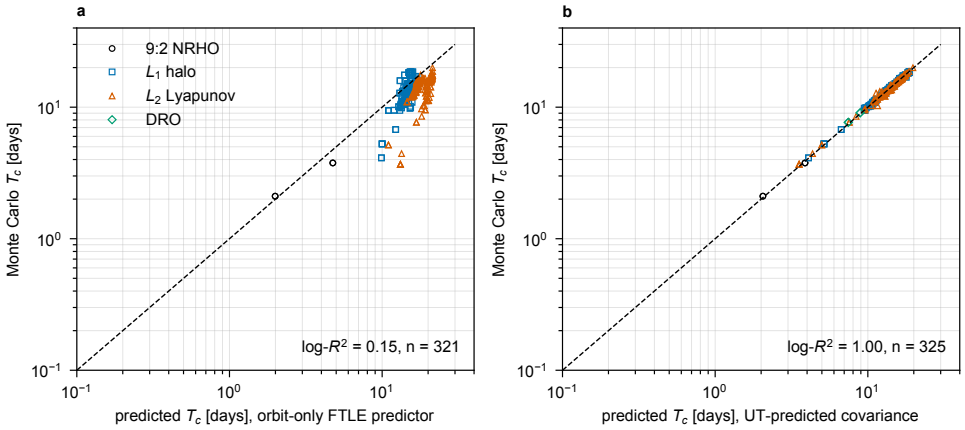


Fig. 10 Monte Carlo custody horizon (ten fields) against **a** the orbit-only FTLE predictor and **b** the horizon claimed by the UT-propagated covariance. The dashed line is equality

−0.25 and 0.04 for the three search sizes. The *covariance-based* predictors succeed: the UT predictor reaches log-R^2 = 0.96, 1.00 and 0.93 for N = 1, 10 and 100 (bootstrap 95 % intervals 0.87–1.00, 0.99–1.00 and 0.85–0.98), and the linear predictor 0.92, 0.99 and 0.97. The FTLE measures stretching along the flow's most unstable direction, but the post-fit covariance is highly anisotropic: angles-only data constrain some directions far better than others, and only the uncertain directions that project onto the sky matter for the search. Predicting custody requires the growth of the *actual* covariance, which the unscented transform supplies from 13 sigma points instead of hundreds of samples.

Linear prediction is adequate for short gaps but breaks down on unstable orbits (Fig. 11). On the L_1 halo, the median linear containment first falls below 95 % at a gap of 9.5 d; on the L_2 Lyapunov orbit it does so only at 18.6 d. After 14 days, the L_1 halo's linear 99 % sky ellipse contains only 5.3 % of the true probability mass (median over all L_1 halo arcs), while the UT ellipse contains 98 %. A search sized and shaped by the linear ellipse would therefore miss most of the target's probability mass. The linear predicted centre, by contrast, generally stays close to the true mean, so the linear failure lies mainly in the predicted spread rather than the predicted position. The UT-centred search gave false custody at N = 10 (a claimed horizon more than 10 % longer than the actual one) in 2 of the 253 distinct arcs with a finite actual horizon: an NRHO arc whose claim exceeds 30 d while the UT-centred circle loses the target at 18.4 d, and an L_2 Lyapunov arc claimed to 11.6 d but lost at 10.2 d. On the unstable orbits, even the UT ellipse loses containment for long gaps: its median containment first falls below 95 % at 18.6–19.5 days and reaches minima of 65–72 %. The recovery near 30 days is an artefact of the small-angle projection once the distribution covers tens of degrees, long after custody is lost.

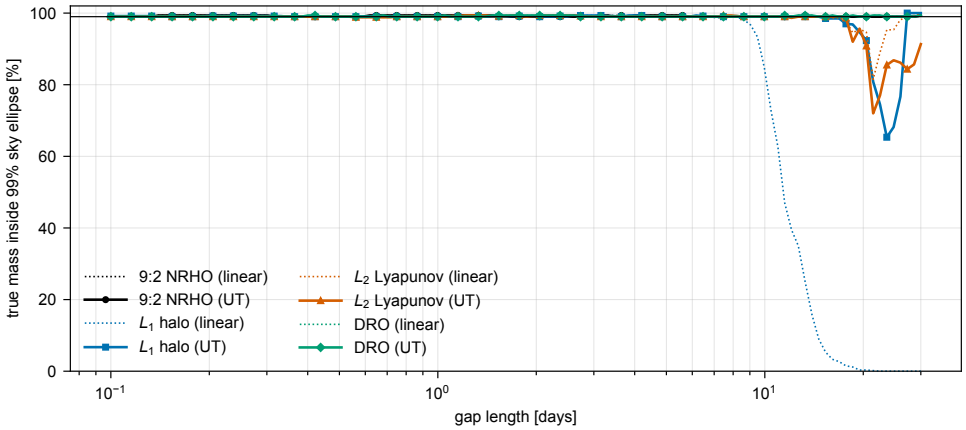


Fig. 11 Share of the true probability mass inside the linear (dotted) and UT (solid, with markers) 99 % sky ellipses, against gap length (median over 7-day arcs). The ideal value is 99 % (black line)

9 Ephemeris Cross-Check

The CR3BP results are checked in an Earth-centred ephemeris model with DE440 Sun and Moon point masses [34] and cannonball solar radiation pressure (area-to-mass ratio $0.01\,\mathrm{m^2\,kg^{-1}}$, $C_r = 1.3$, cylindrical Earth shadow). Each orbit is first transitioned to its ephemeris counterpart by multiple shooting over 45-day windows. Patch points are placed every 0.1 periods and corrected by Levenberg–Marquardt followed by minimum-norm Newton iterations, with STMs from the gravity-gradient variational equations. The L_1 halo, L_2 Lyapunov and DRO counterparts are continuous to better than 0.4 m and stay within 3500–5200 km (median) of their CR3BP shapes. The NRHO counterpart retains a continuity residual of 50–124 km (median to maximum over windows), because the corrector stalls in the ill-conditioned perilune segments; this is a limitation of the present corrector. For 60 custody cases (58 distinct arcs; blackout and phase scenarios, 1- and 7-day arcs), the CRLB covariance is mapped to inertial coordinates and propagated in the ephemeris model.

The conclusions carry over (Figs. 12 and 13, Table 5). The median ratio of ephemeris to CR3BP horizon is 0.93–1.00, and the growth curves overlap. In 5 of the 58 distinct arcs, local differences in the ephemeris dynamics shift T_c by more than 3 days or across the 30-day limit. Blackout survival is unchanged except for one single-case difference in a four-case group (L_1 halo, 1-day arcs). The UT predictor again reaches log-$R^2 = 1.00$ for $N = 10$, but only 0.80 for $N = 1$ (41 finite cases, bootstrap interval 0.37–1.00) and 0.83 for $N = 100$ (30 finite cases, bootstrap interval 0.58–0.99). With so few finite cases these two values are uncertain; at a search radius of 5.6°, moreover, the distribution is too curved for any single Gaussian. Linear false custody persists, and on the L_2 Lyapunov orbit it becomes *worse*. At 14 days the linear ellipse holds 12.0 % of the mass for the L_1 halo (9.9 % for the same arcs in the CR3BP) and 79.9 % for the L_2 Lyapunov orbit (98.7 % in the CR3BP), while the UT ellipse holds 98–99 %.

Table 5 Ephemeris (DE440 + solar radiation pressure) versus CR3BP: median custody horizon (ten fields) and median ratio, by tracking-arc length

Orbit	Arc	n	CR3BP T_c	Ephemeris T_c	Ratio
9:2 NRHO	1 d	8	>30 d	28.3 d	0.98
	7 d	7	>30 d	>30 d	—
L_1 halo	1 d	4	8.4 d	7.8 d	0.93
	7 d	11	15.9 d	15.8 d	1.00
L_2 Lyapunov	1 d	5	11.9 d	11.3 d	0.95
	7 d	10	15.8 d	16.4 d	1.00
DRO	1 d	6	>30 d	>30 d	1.00
	7 d	9	>30 d	>30 d	—

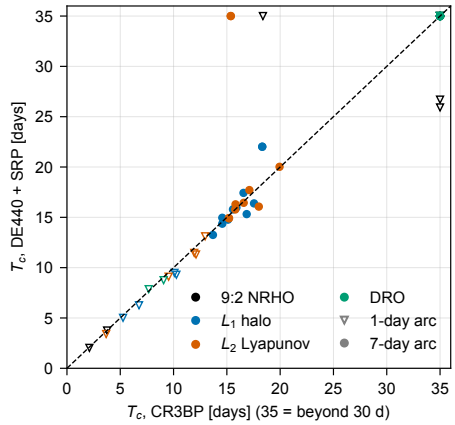


Fig. 12 Custody horizon in the CR3BP and in the ephemeris model (ten fields); open markers: 1-day arcs, filled: 7-day arcs

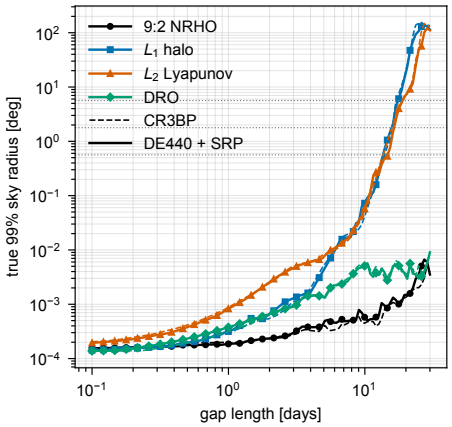


Fig. 13 Median sky-radius growth after 7-day arcs: CR3BP (dashed) and ephemeris (solid). Dotted grey lines are the search radii

10 Horizon Across Orbit Families

The four representative orbits do not show whether the horizon is a property of the particular orbit or of its family's dynamics. The custody computation of Section 4 is therefore repeated for members of all five families of Section 3.1. Up to eight members are taken evenly along each family among those that stay more than 200 km above the lunar surface. Members observable for less than 1 % of the year or without an observable arc are dropped, and 30 members remain (seven L_1 halo, seven L_2 halo and NRHO, six L_1 Lyapunov, five L_2 Lyapunov and five DRO members). Each member starts at a random orbital phase at the epoch. Tracking arcs of 3 and 7 days end at the last visible measurement before each of eight dates 45 days apart, observed by the reference sensors (1 m telescopes, Tri-3+S network, 5° exclusion), which gives 413 cases. The stability index of a member is $\nu = \max|\lambda + 1/\lambda|/2$ over the eigenvalues λ of its monodromy matrix; $\nu = 1$ means linearly stable.

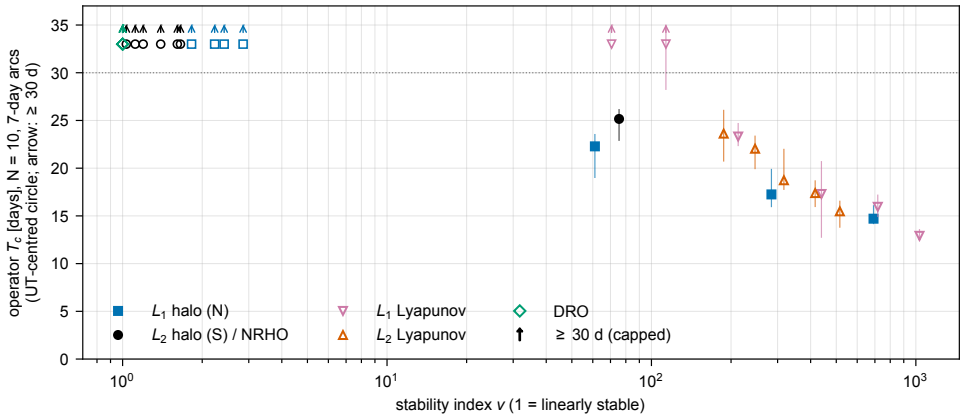


Fig. 14 Operator custody horizon (UT-centred circle, ten fields, 7-day arcs) against the stability index ν for 30 members of five families: median over arcs (markers) and 10–90 % range over arcs (vertical lines). Members whose median exceeds 30 d are drawn above the dotted line with an arrow. Open markers: Lyapunov families and members beyond 30 d

10.1 Stability index

Figure 14 shows the operator horizon against ν. After 7-day arcs, all 15 members with $\nu < 3$ (every DRO, six of the seven L_2 halo and NRHO members and four L_1 halo members) keep custody beyond 30 d (member median). After 3-day arcs all but two of them do; the exceptions are two L_1 halo members with $\nu = 2.4$ and 2.9, lost at 29.1 and 26.9 d. Above $\nu = 10$ the horizon falls as ν grows: after 7-day arcs the finite member medians span 12.9–25.2 d, and two L_1 Lyapunov members ($\nu = 71$ and 113) stay beyond 30 d. For the 15 members with $\nu > 10$, the Spearman rank correlation between the member median horizon and $\log \nu$ is $\rho = -0.92$ (3-day arcs) and -0.91 (7-day arcs), both with $p < 10^{-5}$; horizons beyond 30 d enter as ties at 30 d. Including the nearly stable members (all 25 members with $\nu > 1.01$; the DROs have $\nu = 1.00$) gives -0.97 and -0.91.

Within families, the correlation is strong for the L_1 halo and both Lyapunov families (ρ between -0.91 and -1.00 for both arc lengths, with five to seven members each). The L_2 halo family is the exception, with $\rho = -0.61$ ($p = 0.14$) for both arc lengths. Six of its seven members are nearly stable ($\nu \leq 1.7$) and keep custody beyond 30 d, so only one member ($\nu = 75$) carries ranking information, and this family on its own neither supports nor contradicts the trend.

The stability index therefore separates orbits that keep custody for a month from orbits that do not, and it orders the unstable ones. It does not predict individual cases: at a given ν the horizon varies by several days from arc to arc (vertical lines in Fig. 14). This agrees with Section 8. The index ranks members by their median horizon over a year, whereas the horizon after a particular arc is set by that arc's post-fit covariance, which no orbit-only quantity such as the FTLE can supply.

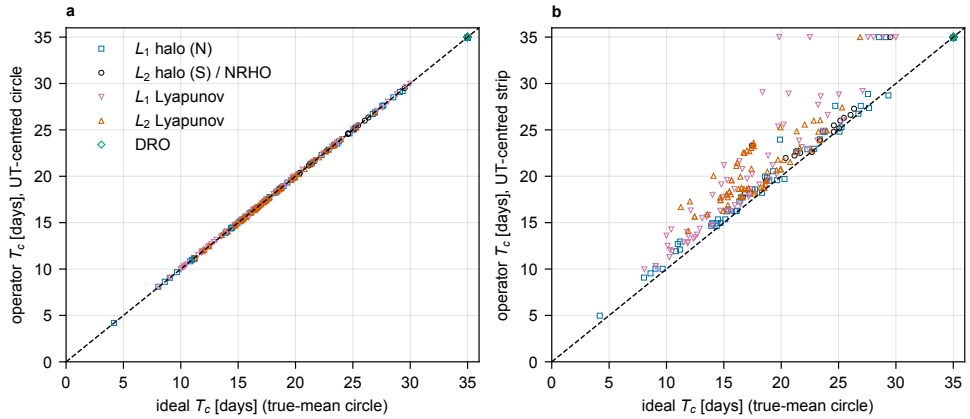


Fig. 15 Operator against ideal custody horizon (ten fields) for the 413 family-sweep cases: **a** circle centred on the UT prediction, **b** strip along the UT-predicted sky ellipse. Horizons beyond 30 d are plotted at 35 d; the dashed line is equality

10.2 Operator and strip searches

Over all 413 cases, the operator horizon differs from the ideal one by at most 0.17 d for $N = 1$, 0.38 d for $N = 10$ and 1.18 d for $N = 100$, and the median ratio of the two is 1.00 for all three search sizes (Fig. 15a). The result of Section 7 therefore holds across the families: centring the search on the unscented prediction costs no custody time.

An operator can also exploit the elongation of the predicted uncertainty by laying out the N fields as a strip along the major axis of the UT-predicted sky ellipse, with the aspect ratio of the ellipse, at most 10° long, and taking the worst of orientation errors of ±2°. Relative to the ideal circle, the median horizon of this strip is 0.97, 1.08 and 0.98 for $N = 1$, 10 and 100 (Fig. 15b). At $N = 10$, the strip gains more than 3 d in 54 of the 206 cases with a finite ideal horizon, 49 of them on the Lyapunov families. These are the cases whose true distribution is thinnest across the strip: where the circle loses custody, the 99 % cross-track half-width of the distribution is 0.6″ (median), against 1.5″ for the other cases. Such thin distributions follow from a Cramér–Rao covariance without measurement biases, timing errors or force-model errors. The strip gain is therefore reported as a possible improvement that remains to be confirmed with a more realistic covariance, not as a result.

11 Robustness to Unmodelled Errors

The horizons so far start from the Cramér–Rao bound, which assumes that white measurement noise is the only error. Real astrometry also has errors that do not average down with the number of measurements. They are added here as consider parameters [25]: quantities that the estimator does not solve for, but whose uncertainty enters the error of its estimate. The measurements become $\boldsymbol{z}_k = h(\boldsymbol{x}_k) + \boldsymbol{b}_s + \boldsymbol{b}_c + \boldsymbol{v}_k$, with a constant angle bias $\boldsymbol{b}_s$ per site and sky axis, a common bias $\boldsymbol{b}_c$ shared by all sites (for example catalogue zonal errors or a common reduction pipeline), and a constant

clock offset per site. The dynamics get an error in the area-to-mass ratio A/m of a cannonball solar radiation pressure model ($C_r = 1.3$), which acts during the arc and keeps acting during the gap. Two cases are distinguished: the operator models solar radiation pressure but A/m is uncertain by 30 % (for A/m from 0.005 to 0.05 $\mathrm{m^2\,kg^{-1}}$), or the operator does not model it at all, so that the truth carries the full A/m. With S the sensitivity of the least-squares estimate to the consider parameters, mapped to the arc end, and C their covariance, the arc-end covariance becomes $P = P_{\mathrm{CRLB}} + SCS^{\top}$. This covariance, together with the solar-pressure error during the gap, defines the Monte Carlo truth; the operator still plans with the Cramér–Rao covariance and the unperturbed CR3BP, as in Section 7. The nominal case combines a 0.1″ bias per site, a 1 ms clock offset and $A/m = 0.01 \pm 30\,\%$; the conservative case combines 0.5″ per site, a 0.1″ common bias, 10 ms and $A/m = 0.05 \pm 30\,\%$. Both are also run with 0.3″ instead of 1″ white noise. The study uses the phase arcs every 30 days (126 arcs of 1, 3 and 7 days); without the additional errors it reproduces the horizons of Section 7 exactly.

The horizons hold (Fig. 16). For a ten-field search and the 67 arcs whose reference horizon is below 30 d, the nominal errors shorten the operator horizon by a median of 1 % (ratio 0.99, 10th percentile 0.96). A per-site bias costs little up to 0.2″ (ratio 0.99) and 5 % at 0.5″ (ratio 0.95, with 5 arcs losing more than 20 %). A common bias matters less than independent site biases, and clock offsets have no effect: the targets move across the sky at a median of $0.4''\,\mathrm{s^{-1}}$ and, in 95 % of the measurements, at less than $0.7''\,\mathrm{s^{-1}}$, so 10 ms shifts them by a few milliarcseconds. With solar radiation pressure modelled, the median ratio stays at 0.98–1.00 up to $A/m = 0.05\,\mathrm{m^2\,kg^{-1}}$, although the 10th percentile falls to 0.82. Not modelling solar radiation pressure at all is the largest single effect: at $A/m = 0.05\,\mathrm{m^2\,kg^{-1}}$ the median ratio is 0.94 and 8 arcs lose more than 20 %. It is also the only error that moves the centre of the search: the truth drifts away from the prediction, so the operator horizon falls while the ideal horizon hardly changes (ratio 0.99). The conservative case gives a median ratio of 0.93 (10th percentile 0.71, 13 arcs losing more than 20 %). The near-stable orbits are barely affected: at most 2 of the 28 NRHO arcs fall below 30 d, and no DRO arc does.

Systematic errors matter more when the white noise is smaller. With 0.3″ noise the horizons themselves are longer, by a median of 1.8 d, but the nominal errors then cost 4 % (ratio 0.96) and the conservative errors 19 % (ratio 0.81, with 46 % of the arcs losing more than 20 %), because the biases dominate the post-fit covariance.

What does not hold is the Cramér–Rao claim. An operator who sizes the search from the Cramér–Rao covariance claims a horizon more than 10 % longer than the actual one in 1.5 % of the arcs without the additional errors, in 9 % under the nominal errors, in 34 % with 0.5″ site biases, and in 82 % for the conservative errors with 0.3″ noise. The search should therefore be sized from a covariance inflated for these errors, such as the consider covariance used here.

12 Discussion

12.1 Operational implications

Six rules follow from the results.

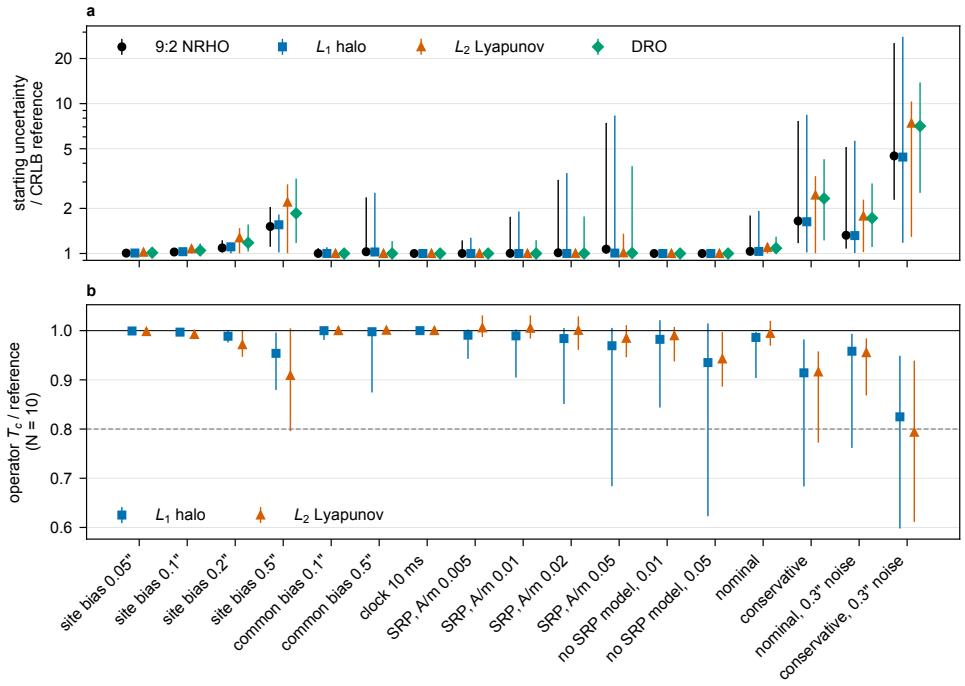


Fig. 16 Effect of unmodelled errors on custody: **a** starting uncertainty relative to the Cramér–Rao reference (largest 1σ position axis), **b** operator horizon (ten fields) relative to the reference for the arcs whose reference horizon is below 30 d. Markers are medians and lines 10–90 % ranges. Configurations: per-site bias, common bias, clock offset, solar radiation pressure modelled with A/m uncertain by 30 %, solar radiation pressure not modelled, and the nominal and conservative combinations, the last two also with $0.3''$ white noise (reference: the $0.3''$ Cramér–Rao bound). An unmodelled force shifts the mean rather than widening the covariance, so its value in **a** is 1. The dashed line marks a 20 % loss

1. Sensor design for cislunar tracking from the ground should minimise the usable Moon-exclusion angle (baffling, scattered-light rejection) before increasing aperture.
2. The monthly lunar-phase blackout cannot be closed by adding ground sites. Custody through it depends on the post-fit covariance, so tasking should concentrate at least 3 and preferably 7 days of tracking before each predicted blackout.
3. Search planning after a gap should use unscented (or Gaussian-mixture) prediction; in the CR3BP, linear prediction on the L_1 halo is safe only for gaps shorter than about 10 days, and in the ephemeris model it also fails on the L_2 Lyapunov orbit by 14 days (Section 9).
4. Reacquisition after a long gap should use a Gaussian-mixture filter, which costs about twice as much as a UKF; it was the only filter to end consistent in at least 99 % of the L_1 halo reacquisition runs.
5. On the NRHO, a narrow (single-field) search should be scheduled away from perilune, where the sky-plane uncertainty briefly swells before shrinking again (Section 4).

6. The search size should come from a covariance inflated for site biases and solar-pressure errors, not from the Cramér–Rao bound: the horizons are robust to these errors, but Cramér–Rao claims of custody become false in a third of the arcs once site biases reach 0.5″ (Section 11). Solar radiation pressure should be modelled.

Because the operator horizon matches the ideal one, these horizons are achievable by an operator who centres the search on a UKF prediction. The stability index gives a first screening of a catalogue: in this study, members with $\nu < 3$ kept custody beyond 30 days after 7-day arcs, while members with $\nu > 10$ needed re-observation within about 13–25 days (Section 10). For the individual arc, the UT predictor tracks the Monte Carlo horizon closely for $N \leq 10$ and could be evaluated inside a tasking scheduler for every catalogued object at negligible cost.

12.2 Limitations

- This is a simulation study; no real observations are processed.
- The post-fit covariance is the Cramér–Rao bound, which the filters reach in tracking but not always immediately after a gap. Section 11 adds site biases, clock offsets and solar-pressure errors for 126 arcs of the four representative orbits; the family sweep and the strip gains of Section 10, which rest on very thin distributions, were not re-tested with them.
- Sensor parameters (limiting magnitudes, extinction, dark-sky brightness) are nominal. The moonlight model of Krisciunas and Schaefer [35] is extrapolated below 10° from the Moon, and the exclusion angle stands in for instrument-specific scattered light.
- Targets are passive, non-manoeuvring spheres of 2 m diameter, and data association and clutter are not modelled.
- Each 99 % quantile is estimated from 300 samples. Re-running 17 representative arcs with 2000 samples changes T_c by a median of 0.08–0.13 d (1–2 %) and by at most 0.7 d for $N = 10$ and 100. At $N = 1$, one NRHO arc whose 99 % radius only just reaches the search radius during a perilune passage moves from 5.3 to 17.6 d.
- The NRHO ephemeris counterpart is not fully continuous.
- The CR3BP study places orbits in the real Earth–Moon direction with analytic ephemerides, and the DE440 check covers 60 of the 523 cases.

12.3 Future work

The main next step is validation with real observations, by proposing observing requests at Indian facilities (the 2 m Himalayan Chandra Telescope at Hanle, the telescopes of the Aryabhatta Research Institute of Observational Sciences at Devasthal, and the 0.7 m GROWTH-India robotic telescope) for targets and nights selected from the custody maps. Other extensions are custody-aware sensor tasking that uses the UT-predicted T_c, mixed ground and space-based networks, and manoeuvre detection.

13 Conclusions

The custody horizon was defined for ground-based, angles-only tracking of cislunar periodic orbits and mapped over a year of physically generated observations, for a search centred on the operator's own unscented prediction. This operator horizon equals that of an idealised search centred on the true mean (median ratio 1.00), so the horizons reported here can be achieved in practice. The targets studied stay within 14° of the Moon, so the exclusion angle and the lunar phase control visibility, and monthly blackouts of up to two weeks are unavoidable from the ground. The near-stable 9:2 NRHO and DRO keep custody for more than 30 days after three or more days of tracking. The unstable L_1 halo and L_2 Lyapunov orbits keep it for 11–17 days (median, ten fields), with the tracking-arc length before a gap as the controlling factor. Across 30 members of five orbit families, nearly stable members keep custody beyond 30 days and, above a stability index of 10, the horizon falls with the index (Spearman $\rho = -0.92$ and -0.91 for 3- and 7-day arcs). The horizon after an individual arc is predicted closely by the unscented transform of the post-fit covariance, but not by the orbit's FTLE alone, and linear prediction gives false custody on unstable orbits after long gaps. Realistic site biases, clock offsets and solar-pressure errors shorten the horizons by about 1 % at nominal levels and by 7 % (median) in a conservative case, but make custody claims based on the Cramér–Rao covariance overconfident, so the search should be sized from an inflated covariance. After a 3-day gap on the L_1 halo, a Gaussian-mixture UKF reacquired in all 100 runs, against 37 % for the EKF and 77 % for the UKF. The main conclusions are reproduced, for 60 of the 523 cases, in a DE440 ephemeris model with solar radiation pressure.

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Declarations

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Data and code availability. The code (CR3BP and ephemeris dynamics, sensor model, estimators, custody analysis and unit tests), the scripts that regenerate every figure and table, the generated data, and a script that checks every number in this paper against those data are available in the `cislunar-custody` repository (https://github.com/Bishwaswarup/cislunar-custody) and archived at Zenodo (https://doi.org/10.5281/zenodo.23075048); release v1.1.0 corresponds to this paper.

Author contributions. B.N. conceived the study, developed the software, performed the analysis and wrote the manuscript.

Ethics approval and consent to participate. Not applicable.

Table A1 Reference parameters of the simulations

Quantity	Value
CR3BP	$\mu = 0.0121505842$, length unit 384 400 km, time unit 4.3425 d
Epoch, grid	2027-01-01 00:00 UTC, 10 min, one year
Sites	Hanle (32.78°N, 78.96°E, 4.5 km); La Palma (28.76°N, 17.88°W, 2.4 km); Haleakala (20.71°N, 156.26°W, 3.05 km); Siding Spring (31.27°S, 149.06°E, 1.16 km)
Constraints	Sun $< -18°$; elevation $> 20°$; exclusion 2/5/10° (reference 5°); no occultation; sunlit
Telescopes	limiting magnitude 19.5 / 21.5 / 23.0 for 0.36 / 1 / 2 m (reference 1 m)
Target	Lambertian sphere, radius 1 m, albedo 0.2
Measurements	topocentric right ascension and declination, $\sigma = 1''$, every 10 min per site when visible
Filters	prior 1000 km / 10 m s^{-1}; process-noise PSD 10^{-18} km^2 s^{-3}; UKF $\alpha = 1$, $\beta = 2$, $\kappa = 0$; GM split threshold 0.02 nats, $\leq$40 components; PF 3000 particles, handover at 50 km; tracking 20 runs, reacquisition 100 runs
CRLB	weak prior 10^5 km, 1 km s^{-1}
Custody	fields of 1°, $N \in \{1, 10, 100\}$; 300 Monte Carlo samples; 120 gap lengths from 0.1 to 30 d
Ephemeris	DE440 Sun and Moon, SRP $A/m = 0.01$ m^2 kg^{-1}, $C_r = 1.3$
Family sweep	30 members of five families, random orbital phase; 3- and 7-day arcs ending before eight dates 45 d apart; 413 cases; strip at most 10° long, orientation error ±2°

Appendix A Reference Parameters

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